

Electromagnetic Induction

Changing magnetic fields make electric fields: the principle behind every generator, transformer, induction cooker and wireless charger.

By **Pratik Bhanuse**, PhD Physics candidate, Western University · pratikbhanuse.github.io

Ørsted showed that currents make magnetic fields. In 1831 Michael Faraday found the reverse: a **changing** magnetic field makes an electric current. This discovery of **electromagnetic induction** made electrical power practical. Almost all the world's electricity comes from generators that spin coils in magnetic fields, and transformers step its voltage up for transmission and down for use. Induction also powers wireless chargers, induction hobs and metal detectors, and it governs the safety of MRI scanners and the operation of brain-stimulation devices. This chapter develops Faraday's and Lenz's laws, motional EMF, induced electric fields, eddy currents, inductance, and the energy stored in magnetic fields.

YOU WILL BE ABLE TO

- Calculate magnetic flux and apply Faraday's law of induction.
- Use Lenz's law to find the direction of induced currents, and relate it to energy conservation.
- Analyse motional EMF in sliding rods and rotating coils (generators), including the forces and power involved.
- Describe induced, non-conservative electric fields and eddy currents.
- Calculate self- and mutual inductance, and analyse transformers.
- Find the energy stored in inductors and the energy density of magnetic fields.
- Solve RL circuits using the time constant L/R .

THINK FIRST

Drop a strong magnet down a vertical copper pipe and it falls remarkably slowly, taking several seconds to emerge from a 1 m pipe. Copper is not magnetic, so a magnet does not stick to it. What slows the magnet down?

Induced currents. As the magnet falls, the magnetic flux through each ring of the pipe below it increases, and the flux through each ring above it decreases. By Faraday's law these changes induce circulating currents (*eddy currents*) in the copper. By Lenz's law, the currents flow in directions that **oppose the change**: the ring below repels the approaching magnet, and the ring above attracts it from behind. The magnet quickly reaches a slow terminal speed at which its weight is balanced by this magnetic drag. Its lost gravitational energy becomes heat in the copper. The same principle provides braking in trains and roller coasters, with no contact and no wear.

§20.1

Magnetic flux

The **magnetic flux** through a surface measures how much magnetic field passes through it:

$$\Phi_B = \int \vec{B} \cdot d\vec{A}, \quad \text{or} \quad \Phi_B = BA \cos \theta \text{ for a uniform field and a flat surface.}$$

The SI unit is the **weber**: $1 \text{ Wb} = 1 \text{ T m}^2$. The flux through a loop can change in three ways: the field changes, the area changes, or the orientation changes.

§20.2

Faraday's law and Lenz's law

KEY IDEA

Faraday's law. The EMF induced in a loop equals the rate of change of the magnetic flux through it. For a coil of N turns:

$$\mathcal{E} = -N \frac{d\Phi_B}{dt}.$$

Lenz's law (the minus sign). The induced current flows in the direction whose own magnetic field **opposes the change in flux** that produced it.

Lenz's law is required by **energy conservation**. Suppose the induced current *aided* the change. Pushing a magnet towards a coil would then induce a current that pulled the magnet in faster, which would induce more current, and so on: a perpetual motion machine. Instead, you must always do work against the induced currents, and that work becomes the electrical energy delivered.

Applying Lenz's law:

1. Find the direction of $\vec{B}$ through the loop, and whether the flux is increasing or decreasing.
2. The induced field opposes the *change*. If the flux is increasing, the induced field points against $\vec{B}$. If it is decreasing, the induced field points along $\vec{B}$.
3. Use the right-hand grip rule to find the current direction that produces that induced field.

WORKED EXAMPLE 20.1

A coil in a changing field

A 200-turn coil of area 50 cm^2 sits with its plane perpendicular to a uniform field. The field rises steadily from 0 to 0.50 T in 0.10 s . Find the induced EMF. If the coil's resistance is 2.0Ω , find the current and the total charge that flows.

Solution

$$\frac{d\Phi_B}{dt} = A \frac{dB}{dt} = (5.0 \times 10^{-3}) \frac{0.50}{0.10} = 0.025 \text{ Wb/s}, \quad |\mathcal{E}| = N \frac{d\Phi_B}{dt} = 200(0.025) = 5.0 \text{ V}$$

$$I = \frac{\mathcal{E}}{R} = 2.5 \text{ A}, \quad Q = It = 0.25 \text{ C}.$$

Note. The total charge is $Q = N \Delta\Phi/R$, which does not depend on *how fast* the flux changes. This is the principle of the *search coil* and the *fluxmeter*, which measure flux changes by measuring the charge that flows.

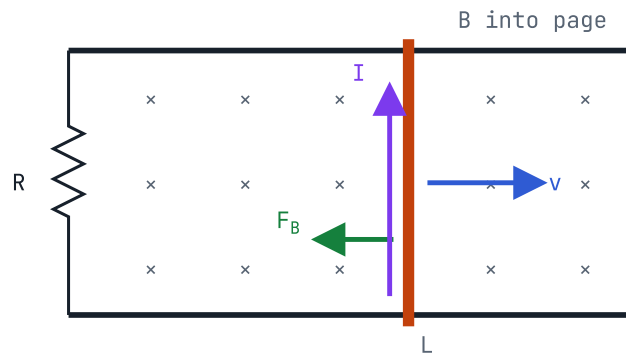
§20.3

Motional EMF

When a conductor moves through a magnetic field, the free charges inside it move with it and feel the force $q\vec{v} \times \vec{B}$. The charges are pushed along the conductor, producing an EMF without any change in $\vec{B}$ at all.

A sliding rod. A conducting rod of length L slides at speed v along two rails connected by a resistor R , in a uniform field B perpendicular to the plane of the rails. The circuit encloses a growing area Lx , so the flux BLx increases at the rate BLv :

$$\mathcal{E} = BLv, \quad I = \frac{BLv}{R}.$$



A rod sliding on rails in a magnetic field into the page. The flux through the circuit increases, so an EMF BLv drives an anticlockwise current. The magnetic force on the current in the rod opposes the motion (Lenz's law).

Forces and energy. The current I in the rod feels a force ILB that **opposes** its motion, which is Lenz's law again. To keep the rod moving at constant speed, an external agent must push with $F = ILB = B^2L^2v/R$. The mechanical power it supplies,

$$Fv = \frac{B^2L^2v^2}{R},$$

equals the electrical power dissipated in the resistor, $I^2 R = \mathcal{E}^2 / R$. Mechanical energy is converted into electrical energy and then into heat. This is the essence of a generator.

WORKED EXAMPLE 20.2

A sliding-rod generator

In the figure, $L = 0.50$ m, $B = 0.40$ T, $v = 2.0$ m/s and $R = 0.80 \Omega$. Find the EMF, the current, the force needed to keep the rod moving, and the power, and check that energy is conserved.

Solution

$$\begin{aligned}\mathcal{E} &= BLv = (0.40)(0.50)(2.0) = 0.40 \text{ V}, & I &= \frac{0.40}{0.80} = 0.50 \text{ A.} \\ F &= ILB = (0.50)(0.50)(0.40) = 0.10 \text{ N}, & P_{\text{mech}} &= Fv = 0.20 \text{ W.} \\ P_{\text{elec}} &= I^2 R = (0.50)^2(0.80) = 0.20 \text{ W. } \checkmark\end{aligned}$$

The AC generator

A coil of N turns and area A rotating at angular speed ω in a uniform field B has a flux $\Phi = BA \cos \omega t$ through each turn. The induced EMF is

$$\mathcal{E} = -N \frac{d\Phi}{dt} = NBA\omega \sin \omega t = \mathcal{E}_0 \sin \omega t, \quad \mathcal{E}_0 = NBA\omega.$$

This is a sinusoidal alternating EMF, the origin of AC mains electricity (Chapter 21). The faster the coil turns, the larger the peak EMF. As current is drawn, Lenz's law produces a back-torque on the coil, so the turbine must work harder. That is why power stations burn more fuel when demand rises.

WORKED EXAMPLE 20.3

A simple generator

A generator coil has 100 turns of area 0.020 m^2 and turns at 3000 rpm in a 0.50 T field. Find the peak EMF and the frequency of the output.

Solution

$$\omega = 3000 \times 2\pi/60 = 314 \text{ rad/s, so the frequency is } 50 \text{ Hz.}$$

$$\mathcal{E}_0 = NBA\omega = 100(0.50)(0.020)(314) = 314 \text{ V.}$$

That is close to the peak value of 230 V rms mains ($230\sqrt{2} = 325$ V).

§20.4

Induced electric fields

When a *stationary* loop sits in a *changing* magnetic field, there is no $\vec{v} \times \vec{B}$ force on its charges, yet a current still flows. Something must push them: a changing magnetic field creates an **electric field**. In general form, Faraday's law reads

$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}.$$

This induced electric field exists even with no wire present. Unlike the field of static charges, its field lines form **closed loops**, and it is **non-conservative**: the work it does around a closed path is not zero. So it cannot be described by a potential.

Inside a long solenoid of radius R whose field is changing, symmetry and Faraday's law give circular field lines with

$$E = \frac{r}{2} \frac{dB}{dt} \quad (r < R), \quad E = \frac{R^2}{2r} \frac{dB}{dt} \quad (r > R).$$

WORKED EXAMPLE 20.4

Induced currents in tissue

A TMS coil produces a field at the brain that changes at about 1.3×10^4 T/s. Estimate the induced electric field 2.0 cm from the axis of the field, and the current density in brain tissue of conductivity 0.30 S/m.

Solution

Treat the field as uniform near the axis:

$$E \approx \frac{r}{2} \frac{dB}{dt} = \frac{0.020}{2} (1.3 \times 10^4) = 130 \text{ V/m}, \quad J = \sigma E = 0.30(130) = 39 \text{ A/m}^2.$$

Evaluate. Fields of about 100 V/m are enough to trigger action potentials in cortical neurons. That is how TMS works without any electrodes touching the brain. The same physics sets the safety limits on how fast MRI gradient coils may switch. Changes faster

than about 20–50 T/s can stimulate peripheral nerves, causing twitching and tingling (Problem P20.16).

§20.5

Eddy currents

In solid conductors, changing flux induces swirling **eddy currents**. They dissipate energy as heat and, by Lenz’s law, they oppose motion.

- **Useful:** induction hobs (a rapidly alternating field heats the pan directly), eddy-current brakes on trains and roller coasters, metal detectors and security scanners, sorting aluminium from scrap, and non-destructive testing for cracks.
- **Unwanted:** losses in transformer and motor cores. These are reduced by building the cores from thin, insulated **laminations**, or from powdered ferrites, which break up the current paths.
- **MRI safety:** the scanner’s radio-frequency fields induce currents in the body that deposit heat. The *specific absorption rate* (SAR) is limited to a few W/kg. Metallic implants and loops of wire, such as ECG leads, can concentrate the heating and cause burns, so they are carefully managed.

§20.6

Inductance

Self-inductance

A current I in a coil produces a magnetic flux through the coil’s own turns. If I changes, this self-flux changes and induces an EMF in the coil itself that opposes the change. The **self-inductance** L is defined by

$$N\Phi_B = LI, \quad \mathcal{E} = -L \frac{dI}{dt}.$$

The SI unit is the **henry**: $1 \text{ H} = 1 \text{ Wb/A} = 1 \text{ V s/A}$. An inductor resists *changes* in current, just as mass resists changes in velocity.

A long solenoid with N turns, length ℓ and cross-section A has $B = \mu_0 NI/\ell$, so the flux linkage is $N\Phi = N(\mu_0 NI/\ell)A$ and

$$L = \frac{\mu_0 N^2 A}{\ell} = \mu_0 n^2 A \ell.$$

Inductance grows as the *square* of the number of turns. An iron core multiplies it by μ_r .

Mutual inductance and transformers

When two coils share magnetic flux, a changing current in one induces an EMF in the other:

$\mathcal{E}_2 = -M dI_1/dt$. The **mutual inductance** M is the same in both directions, $M_{12} = M_{21}$ (Problem P20.14).

An **ideal transformer** has two coils wound on a shared iron core, so the same flux passes through every turn of both. Each turn then has the same EMF, $d\Phi/dt$, and so

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}.$$

With no losses, power is conserved: $V_1 I_1 = V_2 I_2$, so $I_2/I_1 = N_1/N_2$. A step-up transformer raises the voltage and lowers the current, and a step-down transformer does the reverse. Transformers only work with **alternating** current, because steady current produces no change in flux.

WORKED EXAMPLE 20.5

Why transmit power at high voltage

A power station sends 100 MW to a city through lines of total resistance 10 Ω . Compare the power lost in the lines if the transmission voltage is (a) 20 kV and (b) 400 kV.

Solution

(a) At 20 kV: $I = P/V = 10^8/(2 \times 10^4) = 5000$ A, so $P_{\text{loss}} = I^2 R = (5000)^2(10) = 250$ MW. That is more than the power being sent, so it is impossible.

(b) At 400 kV: $I = 250$ A, so $P_{\text{loss}} = (250)^2(10) = 0.63$ MW, only 0.6%.

Evaluate. Raising the voltage 20-fold cuts the line losses 400-fold. Transformers make this possible, which is why AC won the “war of the currents” in the 1890s. Today, very long links sometimes use high-voltage DC, converted with power electronics, which has lower losses over very long distances.

IN PRACTICE

Induction in everyday technology and medicine.

- **Wireless charging.** Phones, electric toothbrushes, and medical implants such as cochlear implants, neurostimulators and some pacemakers are powered through the skin by mutual inductance between an external coil and an implanted one, with no wires crossing the skin (Problem P20.12).
- **RFID and contactless cards** draw their power from the reader's alternating field.
- **Residual-current devices (RCDs)** pass the live and neutral wires through a ring core. Normally their currents cancel. Any leakage to earth leaves an unbalanced flux, which induces a signal that trips the breaker (Chapter 17).
- **Electric guitar pickups** sense the vibrating steel strings as changes in flux through a coil wound around a magnet.

§20.7

Energy in an inductor and in the magnetic field

To build up a current I in an inductor, the source must do work against the back EMF $L di/dt$. The power needed is $Li di/dt$, so the total energy stored is

$$U = \int_0^I Li di = \frac{1}{2} LI^2.$$

For a solenoid, $L = \mu_0 n^2 A \ell$ and $B = \mu_0 n I$, so

$$U = \frac{1}{2} \mu_0 n^2 A \ell I^2 = \frac{B^2}{2\mu_0} (A \ell).$$

The energy density of a magnetic field is therefore

$$u_B = \frac{B^2}{2\mu_0},$$

the magnetic counterpart of $u_E = \frac{1}{2} \epsilon_0 E^2$. For the 3 T MRI magnet of Example 19.4, this gave about 3.6 MJ/m³.

§20.8

RL circuits

An inductor in series with a resistor and a battery cannot change its current instantly.

Growth. When the switch is closed at $t = 0$, the loop equation is

$$\mathcal{E} - iR - L \frac{di}{dt} = 0 \quad \Rightarrow \quad i(t) = \frac{\mathcal{E}}{R} \left(1 - e^{-t/\tau} \right), \quad \tau = \frac{L}{R}.$$

Decay. If the battery is removed and the inductor discharges through R :

$$i(t) = I_0 e^{-t/\tau}.$$

This is the same mathematics as the RC circuit of Chapter 17, with $\tau = L/R$ instead of RC . At the instant the switch is closed, an inductor acts like an open circuit: the current starts at zero. After a long time it acts like a plain wire.

WORKED EXAMPLE 20.6

Current growth in an RL circuit

A 0.50 H inductor and a 10 Ω resistor are connected in series to a 12 V battery. Find the time constant, the final current, the current after 0.10 s, and the energy finally stored.

Solution

$$\tau = \frac{L}{R} = \frac{0.50}{10} = 0.050 \text{ s}, \quad I_{\infty} = \frac{12}{10} = 1.2 \text{ A}.$$

$$i(0.10) = 1.2 (1 - e^{-2}) = 1.2(0.865) = 1.04 \text{ A}.$$

$$U = \frac{1}{2} L I^2 = \frac{1}{2} (0.50) (1.2)^2 = 0.36 \text{ J}.$$

COMMON MISTAKE

Never suddenly interrupt the current in an inductor. If a switch opens in a microsecond, $L di/dt$ can reach thousands of volts. The result is a spark across the switch contacts (the principle of a car's ignition coil), and the voltage spike can destroy transistors. Relay and motor coils are therefore fitted with a “flyback” diode, which gives the current somewhere safe to go (Problem P20.18).

CHAPTER SUMMARY

- Flux: $\Phi_B = \int \vec{B} \cdot d\vec{A}$, in webers. Faraday: $\mathcal{E} = -N d\Phi_B/dt$. Lenz: induced currents oppose the change in flux (energy conservation).
- Motional EMF: $\mathcal{E} = BLv$. Rotating coil: $\mathcal{E} = NBA\omega \sin \omega t$.
- Induced electric fields: $\oint \vec{E} \cdot d\vec{l} = -d\Phi_B/dt$. They are non-conservative, with closed field lines, and drive eddy currents.
- Self-inductance: $\mathcal{E} = -L dI/dt$. Solenoid: $L = \mu_0 N^2 A/\ell$. Mutual inductance M . Ideal transformer: $V_2/V_1 = N_2/N_1$ and $I_2/I_1 = N_1/N_2$.
- Energy: $U = \frac{1}{2} LI^2$. Field energy density: $u_B = B^2/2\mu_0$.
- RL circuit: $\tau = L/R$. Growth: $i = (\mathcal{E}/R)(1 - e^{-t/\tau})$. Decay: $i = I_0 e^{-t/\tau}$.

Practice problems

Full step-by-step solutions are in the separate solutions PDF.

LEVEL A – CONCEPT CHECK

P20.1

A strong magnet falls slowly through a vertical copper pipe but quickly through an identical plastic pipe. Explain the difference. Where does the magnet's lost gravitational energy go?

P20.2

(a) Is an EMF induced in a stationary loop in a changing magnetic field? (b) In a loop moving through a uniform, steady field without rotating? (c) In a loop moving into a region of field? Explain each, and say which force (electric or magnetic) drives the charges.

P20.3

Why are transformer cores made of thin, insulated laminations rather than solid iron?

P20.4

When the switch controlling a large electromagnet is opened, a spark often jumps across the switch contacts. Explain why.

LEVEL B – STANDARD PROBLEMS

P20.5

A 50-turn loop of area 0.10 m^2 lies perpendicular to a uniform field that is decreasing at 0.20 T/s . The loop's resistance is 5.0Ω . Find the induced EMF and current. Describe the direction of the current relative to the field.

P20.6

A generator coil has 50 turns of area 0.010 m^2 and rotates at 60 Hz in a 0.20 T field. Find the peak EMF, and write an expression for $\mathcal{E}(t)$.

P20.7

A rod of length 0.40 m and resistance 2.0Ω (the rest of the circuit has negligible resistance) slides on frictionless rails in a 0.50 T field perpendicular to the rails. A constant force of 0.50 N pulls it. Find its terminal velocity, and the power dissipated at that speed.

P20.8

An aircraft with a 60 m wingspan flies horizontally at 250 m/s where the vertical component of the Earth's field is $4.0 \times 10^{-5} \text{ T}$. Find the EMF between its wingtips. Could this be used to power the aircraft's instruments? Explain.

P20.9

A solenoid 25 cm long and 4.0 cm in diameter has 500 turns. Find its inductance, and the energy it stores when it carries 3.0 A .

P20.10

A transformer has 1150 primary turns connected to 230 V mains and 60 secondary turns. Assume it is ideal. Find the secondary voltage. When a 4.0Ω load is connected, find the secondary current, the primary current, and the power transferred.

P20.11

A 2.0 H inductor is connected in series with a 50Ω resistor and a battery. Find the time constant, and the time for the current to reach 90% of its final value.

P20.12

Powering an implant. An external charging coil produces an alternating field of peak value $50 \mu\text{T}$ at 100 kHz at the position of an implanted receiving coil. The implant coil has 30 turns of radius 1.0 cm , with its axis along the field. Find the peak EMF induced in it.

LEVEL C – CHALLENGE PROBLEMS**P20.13**

A square loop of side L , mass m and resistance R falls vertically. Its lower edge has left a region of horizontal uniform field B (perpendicular to the plane of the loop), while its upper edge is still inside. (a) Show that it reaches a terminal speed $v_t = mgR/(B^2 L^2)$. (b) Evaluate v_t for $m = 10 \text{ g}$, $R = 0.050 \Omega$, $L = 0.10 \text{ m}$ and $B = 1.0 \text{ T}$. (c) Explain where the gravitational energy goes at terminal speed.

P20.14

Two coaxial solenoids share the same length ℓ . The inner one has N_1 turns and cross-section A_1 , and the outer one has N_2 turns. (a) Find the mutual inductance by considering the flux through the outer coil due to current in the inner one, and then by the reverse. Show that the two results agree. (b) Evaluate M for $\ell = 0.20 \text{ m}$, $N_1 = 400$, $N_2 = 800$ and $A_1 = 2.0 \text{ cm}^2$.

P20.15

For a long solenoid, show that the energy $\frac{1}{2}LI^2$ equals the integral of $u_B = B^2/2\mu_0$ over the volume inside the solenoid. Interpret the result physically.

P20.16

MRI gradients and nerve stimulation. A magnetic field changes uniformly at a rate dB/dt inside a cylindrical region of radius R . (a) Use $\oint \vec{E} \cdot d\vec{l} = -d\Phi_B/dt$ to find the induced electric field inside and outside the region. (b) An MRI gradient switch produces $dB/dt = 50 \text{ T/s}$ over the torso, which can be modelled as a cylinder of radius 0.20 m . Find the largest induced field. Peripheral nerves can be stimulated by fields of a few V/m. Comment on the scanner's limits on how fast the gradients may switch.

P20.17

Faraday's disc (homopolar generator). A copper disc of radius R rotates at angular speed ω in a uniform field B parallel to its axis. Sliding contacts touch the axle and the rim. (a) Show that the EMF between the axle and the rim is $\frac{1}{2}B\omega R^2$. (b) Evaluate it for $R = 0.50$ m, $B = 1.0$ T and 3000 rpm. (c) Explain why homopolar generators give large currents but low voltages.

P20.18

A relay coil has $L = 10$ H and $R = 24 \Omega$, and is powered from 12 V. (a) Find the steady current and the stored energy. (b) If a switch interrupts the current in 1.0 ms, estimate the induced EMF. (c) A diode is now connected across the coil so that, when the switch opens, the current can circulate through the coil's own resistance. Find the time constant of the decay, and explain how this protects the switch.

All worked examples and practice problems in these notes are original.

Textbooks for further reading and practice, with a chapter-by-chapter map, are listed in the References.

© 2026 Pratik Bhanuse. All rights reserved. · pratikbhanuse.github.io